MIL-E-70610(AR)
1 June 1986
SUPERSEDING
AEI-54C
14 October 1966

MILITARY SPECIFICATION

EIMITE IGNITER

This specification is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This military specification covers Eimite for use in MINE, APERS, M26.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, listed in the solicitation.

SPECIFICATIONS

FEDERAL

RR-S-366	-	Sieve, Test.

MILITARY

MIL-D-98	-	Diphenylamine, Technical
MIL-P-156	-	Potassium Nitrate (KN03).
MIL-N-244	-	Nitrocellulose.
MIL-M-382	-	Magnesium Powder (For Use in Ammunition).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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JAN-M-560 - Magnesium Stearate (Ointment Type).
MIL-B-2427 - Box, Ammunition Packing: Wood, Nailed.
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For.

STANDARDS

FEDERAL

FED-STD-406 - Plastics; Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing.
MIL-STD-650 - Explosive: Sampling, Inspection and Testing.
MIL-STD-1168 - Ammunition Lot Numbering.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8860900 - Drum, Metal, Shipping (Interior)

PUBLICATIONS

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commision Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders should specify "49 CFR 100-199 (latest revision)".)

(Copies of specifications, standards, drawings, publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the

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text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Igniter. The igniter shall comply with all requirements specified on associated drawings and with all requirements specified in applicable specifications.

3.2 Material. The constituent materials used in the manufacture of the Eimite shall comply with the following requirements:

Constituent Material	Conforming to
Nitrocellulose (13.15% N)	Specification MIL-N-244, Grade C, type 1
Potassium Nitrate	Specification MIL-P-156, Class 1
Diphenylamine (added)	Specification MIL-D-98
Magnesium	Specification MIL-M-382, Type III, Granulation No. 16
Boron (See 6.2)	Amorphus - Elemental 90-92%, Moisture content 0.5% max., average particle diameter 1.0 microns, max.

3.3 Composition. The composition of the Eimite shall be as specified in Table I when tested as specified in 4.5.1.

TABLE I

Composition of eimite

Constituent	Percent
Nitrocellulose	40 ± 1
Magnesium	22.6 ± 1.5
Potassium Nitrate	37.4 ± 2
Diphenylamine (added)	1.0 ± 0.2
Boron Elemental MGR (Glaze) (added)	2 ± 1

3.4 Moisture content. The moisture content of the Eimite at the time of loading shall not exceed 0.4 percent when tested as specified in 4.5.2.

3.5 Total volatiles. The total volatiles shall not exceed 0.70 percent when tested as specified in 4.5.3.

3.6 Bulk density. The bulk density shall be 0.45 ± .05 when tested as specified in 4.5.4.

3.7 134.5° Centigrade (C) heat test. The powder shall not change the color of normal methyl violet test paper to a salmon pink

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color in less than 35 minutes, and shall not explode in less than 5 hours when tested as specified in 4.5.5.

3.8 Granulation. The Eimite shall comply with the granulation requirements shown in Table II, using the US Standard sieves conforming to Specification RR-S-366 when tested as specified in 4.5.6.

TABEL II

Granulation requirements

Sieve No.	Percent Retained	Percent Through
40	5 max.	--
100	--	5 max.

3.9 Heat of reaction and the volume of gaseous products. The heat of reaction shall not be less than 1550 cal/gm, and the volume of gaseous products shall be 325 ± 50 ml/gm when tested as specified in 4.5.7.

3.10 First article inspection. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3)

3.11 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Quantity of Eimite inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Quantity of Eimite in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certifiate submitted shall be submstantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Classification of inspections. The inspection requirements specified herein are classifed as follows:

a. First article inspection. (see 4.3)

b. Quality conformance inspection. (see see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below and Table I.

Part Description	Drawings	Quantity
Eimite	See (3.3)	5 lbs.
Drums	8860900	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein. If any items fail to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3.3 Rejection. See MIL-A-48078. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Eimite Igniter				See Below
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Chemical Tests				
	Composition			3.3	4.5.1
	Moisture Content			3.4	4.5.2
	Total Volatiles			3.5	4.5.3
	Bulk Density			3.6	4.5.4
	134°C Heat			3.7	4.5.5
	Granulation			3.8	4.5.6
	Heat of Reaction and Volume of Gaseous Products			3.9	4.5.7
	Drum, Steel (prior to loading) (Dwg. 8860900)				
	Examination for Defects	10		3.1	4.4.2.2
	Sealed Metal Drum (Dwg. 8860900)				
	Examination for Defects	10		3.1	4.4.2.3
	Air Pressure Tests			3.1	5.1.1

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each lot shall consist of Eimite manufactured from one batch which has been subjected to the same processing operations and blended at one time. Each batch shall consist of that quantity of material which has been subjected to the same unit chemical process or physical mixing process intended to make the final product substantially uniform. The lot shall comply with Standard MIL-STD-105 and shall be numbered in accordance with Standard MIL-STD-1168.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-70610(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.1	Eimite	1	1	--
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Composition			3.3(a)	4.5.1
102	Moisture content			3.4(a)	4.5.2
103	Total volatiles			3.5(a)	4.5.3
104	Bulk density			3.6(a)	4.5.4
105	134°C Heat			3.7(a)	4.5.6
106	Granulation			3.8(a)	4.5.7
107	Heat of reaction and volume of gaseous products			3.9(a)	4.5.8
Minor	None defined				

NOTES:

(a) If any sample fails to comply with the applicable requirements, the lot shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-70610(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Drum, Steel (Prior to loading)	1 OF 1	8860900
			NEXT HIGHER ASSEMBLY: --

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Container damaged destroying waterproofness qualities (puntured, cracked, etc.)		0.40%	3.10	Visual
102	Foreign matter on interior		0.40%	3.10	Visual
Minor	None defined				

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-70610(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Sealed Metal Drum				8860900
					NEXT HIGHER ASSEMBLY --
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Head and locking ring assembly improperly assembled (incompletely closed, loose, etc.)		0.65%	3.10	Visual-Manual
102	Gasket missing or broken		0.65%	3.10	Visual
103	Drum damaged (hole, crack, etc.)		0.65%	3.10	Visual
104	Air pressure test (a)			3.1	5.1.1
Minor					
201	Car seal missing, unsealed or improperly positioned		2.5%	3.10	Visual-Manual
202	Marking misleading or unidentifiable including incorrect color code		2.5%	3.10	Visual
203	Painting damaged (bare areas exceed 1/4 square inch)		2.5%	3.10	Visual
204	Poor workmanship		2.5%	3.10	Visual

NOTES: (a) This test shall be performed 100 percent. Drums failing to comply with the specified requirements shall be classed defective and removed from the lot.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1, and the test method paragraph (see 4.5). The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.3 herein.

4.5 Methods of inspection. Five (5) pounds of Eimite shall be selected at random from each lot and subjected to these tests.

4.5.1 Determination of composition.

4.5.1.1 Nitrocellulose. The percentage of nitrocellulose shall be determined in accordance with Standard MIL-STD-286, Method 209.2.1.

4.5.1.2 Potassium nitrate. The percentage of potassium nitrate shall be determined in accordance with Standard MIL-STD-286, Method 310.4.1.

4.5.1.3 Diphenylamine. The percentage of diphenylamine shall be determined in accordance with Standard MIL-STD-286, Method 217.2.

4.5.1.4 Magnesium. A sample (0.3 to 0.5 grams) containing about 0.07 grams of magnesium is accurately weighed into a 250 ml. beaker and digested with 10 ml. of nitric acid and 5 ml. of sulfuric acid on a hot plate until the solution is clear, adding more nitric acid, if needed. The digested solutions are heated until sulfur trioxide is entirely eliminated. When cool, the residue is dissolved in 25 ml. of water and 2 to 3 ml. of hydrochloric acid with slight heating, then transferred to a 400 ml. beaker, adding sufficient water to bring the volume to 125 ml. The magnesium is precipitated, filtered, and ignited in accordance with Specification JAN-M-560, F-4b, starting with ".....add 1.2 gm of ammonium diphosphate." The percentage of magnesium is calculated as follows:

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$$\text{Magnesium, \%} = \frac{21.85A}{W}$$

Where:
A = Weight of precipitate (magnesium pyrophosphate)
W = Weight of sample

4.5.1.5 Boron.

4.5.1.5.1 Special apparatus.

a. pH meter equipped with glass-calomel electrode system, and calibrated with pH7 buffer.
b. Buret, 10 ml., graduated at 0.05 ml. intervals.
c. Hot plate, 11 watts per sq. in., Lindberg Type H-2, or equivalent.

4.5.1.5.2 Special materials.

a. Barium hydroxide solution. Prepare by steeping 3 grams of barium hydroxide octahydrate in a liter of water at least overnight, then filtering through a Whatman No. 12, or equivalent, folded paper.
b. Boric acid, standard solution containing 0.17 mg. of boron per ml. Prepare by dissolving an accurately weighed 0.5 gram portion of boric acid in 500 ml. of water.
c. Hydrochloric acid, 1N solution.
d. Mixed acid. Prepare by stirring 33 ml. of 71% perchloric acid into 12 ml. of water, and into this, stirring 55 ml. of 96% sulfuric acid. This mixture must be permitted to cool before use.
e. Potassium dichromate, 2% solution.
f. Sodium hydroxide, 2N solution, freshly prepared.

4.5.1.5.3 Determination. Accurately weigh a one (1) gram specimen into a 250 ml. erlemeyer flask with standard-taper joint and insert an air-condenser. Through the condenser, add 1 ml. of potassium dichromate solution and, slowly, 10 ml. of mixed acid. Make similar additions to a sample free (blank) flask. (Note 1 following this paragraph) Heat for a half hour at a reflux rate of 3-4 drops per minute (Note 2), then increase the temperature to a rate of 10-15 drops per minute and hold at this temperature until the color of the contents changes from green to orange. Remove the flasks and let them cool to room temperature. Wash the condenser walls and base with water and transfer the contents of the flask to a 250 ml. volumetric flask. Add water to bring the volume to the mark and shake to mix.

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Transfer a 25 mil. aliquot from the volumetric flask to a 150 ml. erlemeyer flask with a standard taper joint. From a 10 ml. Mohr pipet, add 8 ml. of freshly prepared 2N sodium hydroxide solution, a strip of litmus paper (Note 3), and more sodium hydroxide solution, dropwise, until the paper just turns blue. Wash the flask mouth and walls with water. Add 1N hydrochloric acid, a drop at a time, until the color of the litmus paper just reverts to red, and then one drop more. Attach the flask to a water-cooled condenser, reflux for 15 minutes, then remove the flask and lace it in a cold water bath (Note 4).

When cool, wash the contents of the flask into a 150 ml. beaker and insert the electrodes of the pH meter (Note 5). From the 10 ml. buret add barium hydroxide solution (the volume need not be recorded) in increments of about 1 ml. to pH 4, 0.5 ml. between pH 4 and pH 5, 0.25 ml. between pH 5 and pH 6, and 0.1 ml. between pH 6 and pH 6.7. When the pH exceeds 6.7, add only a drop at a time, allowing 30 seconds between drops, until the pH is adjusted to 7.2. Add 5 grams of mannitol and measure the volume of barium hydroxide solution required to restore the Ph to 7.2, adding the solution in progressively smaller increments in the manner already described. Standardize the barium hydroxide solution not earlier than the day before nor later than the day after that on which the titration was performed.

To standardize the barium hydroxide solution, pipet 25 ml. of "blank" solution and 10 ml. of standard boric acid solution into a 150 ml. erlemeyer flask with standard taper joint. Adjust the acidity, reflux, and titrate in the manner described in the previous paragraphs. Calculate the strength of the solution as follows:

$$S = \frac{0.1750 W_1}{V_1} -$$

S = Strength of barium hydroxide solution, mg. of boron per ml.

Where:

V_1 = Volume of barium hydroxide solution required to titrate the aliquot of boric acid, ml.

W_1 = Weight of boric acid in the aliquot of boric acid solution, mg.

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Calculate the boron content of the sample as follows:

$$\text{Boron, \%} = \frac{100V_2S}{W_2}$$

Where:

S = Strength of barium hydroxide solution, mg. of boron per ml. (see previous paragraph).

V_2 = Volume of barium hydroxide solution required to titrate the aliquot of sample solution, ml.

W_2 = Weight of sample represented by the aliquot taken, mg.

Notes:

1. It is not necessary to include a blank with each run of samples. Blank solution from a recent (within a few weeks) run may be used provided that the quantity on hand is sufficient to standardize the barium hydroxide solution.
2. These temperatures may be attained on a Lindberg Type H-2 electric hotplate, first with, and then without, a thickness of 1/8" asbestos board.
3. If the paper should assume the blue shade, wash down the flask and reacidify with 1N hydrochloric acid solution as described subsequently. In later runs, decrease the volume of 2N sodium hydroxide solution added at this point.
4. A vessel containing dry ice in contact with water makes a satisfactory bath.
5. The use of a magnetic stirrer will facilitate subsequent operations. In the event that such a stirrer is used, both housing of the motor and the shielding of the pH meter should be connected to ground.

4.5.2 Determination of moisture content. (see 3.4) The percentage of total moisture shall be determined in accordance with Standard MIL-STD-286, Method 102.1.3.

4.5.3 Determination of total volatiles. (See 3.5) The total volatiles shall be determined in accordance with Standard MIL-STD-286, Method 103.3.3.

4.5.4 Determination of bulk density. (See 3.6) The bulk density shall be determined in accordance with Standard MIL-STD-650, Method 201.3, except that a sample weight of forty (40) grams shall be used.

4.5.5 134.5°C Heat test. (See 3.7) The 134.5°C heat test shall be performed as specified in Standard MIL-STD-286, Method 404.1.2.

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4.5.6 Granulation. (See 3.8) Nest the specified sieve or sieves (see Table II) on a bottom pan. Place a weighed portion of 100 gm of the sample on the upper sieve, cover the sieve, and stroke the assembly for 10 minutes by hand, or for 5 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amount retained or passing through the sieves, and calculate to percentages as required.

4.5.7 Determination of the heat of reaction and the volume of gaseous products. (See 3.9) The determination shall be performed in accordance with the following procedure and using "Parr" Adiabatic Calorimeter, or approved equal.

a. Weigh, aproximately, 3.0 grams of a sample of Eimite to 0.0002 grams in a Coors, size 2, porcelain crucible.
b. Coil 9" of 0.01" diameter platinum wire around a piece of asbestos, 3/16" wide by 3/4" long.
c. Attach ends of wire to the bomb electrodes and place the coil in contact with the surface of the sample and assemble the bomb.
d. Fill the bomb with purified helium to 20 atmospheres and exhaust the helium slowly. Repeat this operation three times. (The final pressure of the bomb is 1 atmosphere.)
e. Fill the bucket with the same quantity of water and at the approximate temperature as in the standardization procedure using the prepared tare as the counterbalance.
f. Place the bucket of water in the calorimeter, set the bomb in the bucket, and attach the electrical connection to the bomb firing terminal.
g. Close the calorimeter with the therometers in place.
h. Adjust the jacket temperature to equal the temperature of the water in the bucket by injecting hot or cold water, as required.
i. Maintain the equalization temperature for a period of five minutes after equilibrium has been reached and record temperature of bucket.
j. Ignite sample.
k. Measure the energy required for ignition. (A 12 volt battery for ignition and a Weston Watt-Second Integrator, Model #808, for determining the ignition energy have been found satisfactory.)
l. Inject hot water into the calorimeter jacket as the temperature of the calorimeter is rising, maintaining equilibrium temperature conditions until the maximum temperature is reached.
m. Record the final temperature of the reaction and ambient air temperature. Check the bomb for leakage, and disassemble the calorimeter.
n. Exhaust the gases from the bomb slowly into the eudiometer and record the volume of gas, the bath and room temperature, and barometric pressure.

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CALCULATIONS:

Determine the volume of the gaseous products of the explosion at standard temperature and pressure (STP) by using the equation shown below:

$$V_o = \frac{V \times 273.1 \times Pc}{(W \times K^o) \times 760}$$

Where:

V_o = Corrected volume in ml. (STP)
V = Uncorrected volume in ml.
P_c = Corrected barometric pressure
W = Weight of sample (gms)
K^o = Absolute temperature of room

o. correct the initial and final temperature according to certificate issued with the calibrated Beckman Thermometer, and include an emergent stem correction.

p. The heat of reaction in cal/gm is calculated from the equation below:

Heat of Reaction =

$$\frac{\text{(Corrected Temp Rise) X (Heat Equivalent) - (Firing Correction)}}{\text{Weight of Sample}}$$

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level C. Five (5) pounds Eimite shall be packaged in a conductive polyolefin bag. (See Note 1) Size of bag shall be 10-1/2 min. X 18 min. Excess air shall be removed from the bag, by hand pressure, prior to twisting end of bag. After twisting end, fold end back on itself and secure with strong elastic band. Place above in Metal Drum MS24347-5 or MS24347D5 Type I or II Drawing F8860900 (Note 2). Seal Drum with locking ring, nut and bolt in accordance with Drawing F8794342. Sealed drum shall withstand an air pressure test of three pounds per square inch for fifteen seconds without leakage.

5.2 Packing.

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5.2.1 Level C. Eight (8) Drums of Eimite shall be packed in a Type I, Class I, Grade C, Spec. MIL-B-2427 wood box. Inside dimensions of wood box to be 34-1/2 X 17-1/4 X 10-1/8. (Note 3) An egg-crate separator manufactured of "B" flute corrugated fiberboard, Domestic class, SW variety, Spec PPP-F-320 shall be used to separate the Drums. The separator in a 4 X 2 pattern shall be a snug fit in the box length and width and be 10 inches high (Note 4). Commercial chipboard fillers 32 X 16 X 1/16 shall be added on top of contents for tight packing. Secure cover of box with 3 straps in accordance with Spec MIL-B-2427.

5.3 Marking.

5.3.1 Drum. Drum shall be marked with tag or with yellow stencil ink. Information to be included are nomenclature, "5 pounds Eimite" and also lot number (if any) and date packed.

5.3.2 Wood box. Box to be marked in accordance with Drawing P8796522. Nomenclature to be: "40 pounds Eimite". DOT (Ref Interstate Commerce Commission Code of Federal Regulations) 49 CFR-71-78 to be: "High Explosives - Dangerous".

Note 1. 8 Mils thick conductive polyolefin, black having surface resistivity not greater than 10^3 ohms per centimeter. Determined in accordance with FED-STD-406, methods 4041 and 4052. Electrostatic bleedoff shall comply with requirements of SPEC-XAS-1152 Para. 4.7.10 and Table I.

SUGGESTED SOURCES:

Custom Materials Inc.
Chelmsford, MA

Crystal - X Corp.
Darby, PA

Note 2. Height (outside of Type I Drum - 9-1/2)
Height (outside of Type II Drum - 10)

Note 3. Inside height of box to be 9-5/8 if Type I Drum is used.

Note 4. Separator to be 9-1/2 inches high if Type I Drum is used.

5.4 Shipping. When drums of Eimite from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Ordering data. See MIL-A-48078. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Ammunition data cards shall be prepared for each lot in accordance with Standard MIL-STD-1167.
c. Level of packaging and packing required.
d. Provisions for submission of First Article samples.

6.2 Boron. A product of American Potash and Chemical Corporation or approved substantial equal.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A365)

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